

Volume

Volume is the measure of three-dimensional space enclosed by a solid object, expressed in cubic units. It answers the question: *how much space does this object occupy?* Volume extends the ideas of length (one dimension) and area (two dimensions) into three dimensions, and it is fundamental to fields ranging from engineering and chemistry to architecture and fluid mechanics.

The most familiar volumes come from standard geometric solids. A rectangular box with length l , width w , and height h has volume $V = lwh$. A cylinder with base radius r and height h has $V = \pi r^2 h$ — essentially the base area multiplied by the height. A sphere of radius r has $V = \frac{4}{3}\pi r^3$, and a cone with the same base and height as a cylinder holds exactly one-third as much: $V = \frac{1}{3}\pi r^2 h$.

Worked example. A water tank is cylindrical with a diameter of 1.2 m and a height of 2.0 m. How many liters does it hold when full?

First, find the radius: $r = 0.6$ m. Then:

$$V = \pi r^2 h = \pi(0.6)^2(2.0) = \pi(0.72) \approx 2.262 \text{ m}^3$$

Since $1 \text{ m}^3 = 1000 \text{ L}$, the tank holds approximately **2262 liters**.

Problem-solving application. Volume calculations often appear in layered or composite situations. Suppose you need to fill a concrete foundation that is a rectangular slab $6 \text{ m} \times 4 \text{ m} \times 0.25 \text{ m}$, but a cylindrical pipe of diameter 0.1 m runs vertically through it. The concrete volume is the slab volume minus the pipe's volume:

$$V_{\text{slab}} = 6 \times 4 \times 0.25 = 6.0 \text{ m}^3$$

$$V_{\text{pipe}} = \pi(0.05)^2(0.25) \approx 0.00196 \text{ m}^3$$

$$V_{\text{concrete}} \approx 6.0 - 0.002 = 5.998 \text{ m}^3$$

The pipe is negligible here, but in precision contexts — pharmaceutical capsules, microchip voids, fuel injectors — such corrections matter enormously.

A practical strategy for any volume problem: identify which standard shape (or combination of shapes) best models the object, look up or derive its formula, substitute values carefully with consistent units, and check the result against physical intuition.